

Mastering Observability

ADVISORY · STRATEGY · PRACTICE

FRAMEWORK EBOOK · METRICS & MAYHEM

The Outcome Question: can your team explain why?

When the number your CFO watches moves, can your team explain why within five minutes? One question, a twenty-minute test to find your own gap, and what to do on Monday.

Allan Mann · Author, Metrics & Mayhem · A working companion to the book

FREE FRAMEWORK EBOOK · SOURCES ARE CHAPTER 2 OF THE BOOK, A PUBLIC POST-MORTEM, A PUBLIC ESSAY AND THE AUTHOR'S OWN EXPERIENCE · WORKED EXAMPLES ARE ILLUSTRATIONS AND SAY SO

AT A GLANCE

When the number your CFO watches moves, can your team explain why within five minutes?

01 **Logs, metrics and traces only describe inputs**
None of them tells you whether the customer finished what they came to do.

02 **Pair one outcome with the signals that move it**
Then put a person's name, not a team's, on the join between them.

03 **Estate-wide baselines miss the quiet services**
In one fifty-service estate, a failing service went unnoticed for more than two weeks.

04 **Your status view can fail with the system**
Early in the 2017 S3 outage, AWS could not update its own health dashboard.

05 **Twenty minutes will find your gap**
The five-minute test needs one outcome and a pen, not a new tool.

01 The gap
02 The practice
03 The evidence
CLOSE On Monday

HOW THIS WAS SOURCED

Chapter 2 of Metrics & Mayhem, AWS's public post-mortem of the 2017 S3 outage, a public essay by Charity Majors, and the author's own experience, with details changed so no organisation can be identified. No vendor surveys. The worked examples are illustrations, and are labelled as such.

PART ONE · THE SCENE

Both sides of the review were right

A quarterly business review. Engineering presents first. Uptime for the quarter: 99.97 per cent. Green across the board, every service level met, and the VP of Engineering is quietly pleased.

99.97%

UPTIME, IN THE SAME QUARTER CHECKOUT-FAILURE TICKETS ROSE 12 PER CENT

Source: illustrative scene, not a real organisation

Then Customer Success stands up. Checkout-failure tickets are up 12 per cent on the quarter. Satisfaction scores for the purchase flow are down. Three enterprise accounts raised it at their last contract review.

Both presentations are accurate. Neither can explain the other's number.

> FROM THE FIELD
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> I have watched every dashboard stay green while the people using a service could not get through it. The complaints came from customers, not from our alerting. The service was low-volume, and our anomaly detection set its baseline across the whole estate, so a small service failing barely moved the needle against the big ones. We tracked it down by hand, and only then could we see where our monitoring gaps were.

That gap, between the system being up and the customer getting through, is what this short book is about. One question closes it, and you can ask it on Monday.

THE QUESTION

One question separates observability from monitoring

When the number your CFO watches moves, can your team explain why within five minutes?

Not "is the system healthy". Not "are the dashboards green". Whether a movement in something the business cares about can be traced, quickly, to the technical cause underneath it.

If the answer is yes, you have observability in the sense that matters to you. If the answer is no, or "it depends who is on call", you have monitoring, and a gap where your explanation should be.

INPUTS AND OUTCOMES

Logs, metrics and traces only describe inputs

If you have been briefed on observability, you have heard about the three pillars: logs, metrics and traces. Logs tell you what happened, metrics tell you how much, and traces tell you where a request spent its time.

Within minutes, someone can say out loud what is happening to the customer, in words the product lead understands, and which part of the estate is responsible. The owner of the join is in the conversation from the start, not found halfway through. The finance lead hears "completion is down four points because the processor is timing out", and knows what that costs.

None of that requires the organisation to see everything. It requires one outcome to be explicable, quickly, by the people who get the call. Start there.

OUTSIDE COMMERCE

The same pairing works without a checkout

Not every organisation sells through a checkout. The pairing works the same way wherever there is an outcome someone outside engineering would recognise.

Take a business that serves partners through an API. The outcome the commercial team watches is the share of partner requests answered within the time the contract promises. Miss it for a week and it shows up in a renewal conversation, long before it shows up in an engineering review.

The technical signals that can move it are different from the checkout case, and just as specific. Response time at the 99th percentile for each partner, not averaged across all of them. Error rates by partner. Authentication and token failures. Requests rejected by rate limits. Latency from the upstream services your API calls in turn. Configuration and deployment changes in the window before the number moved.

Notice the first one. Averaged across every partner, response time can look perfectly healthy while your largest partner has a miserable week, because their traffic is a small share of the total. It is the same trap as a baseline learned across a whole estate. The pairing has to be written at the level the business feels the pain, which is usually one customer, one partner or one journey, not the average.

Chapter 2 of the book names other outcomes that pair just as well: search relevance, content delivery latency, or API response time for key partners.[2] The method does not change. One outcome, the signals that can move it, and a person who owns the join.

THE FIVE-MINUTE TEST

Twenty minutes with a pen will find your gap

Do this on one outcome, not ten. Twenty minutes, pen in hand. If the last answer is more than five minutes, or you had to ask someone to find out, you have found your gap.

1 OUTCOME
Write down the one business outcome your CFO would recognise.

2 SIGNALS
List every technical signal that can move it.

3 VISIBILITY
Mark each signal your team can see today. Leave the rest blank.

PART THREE · HOW BIG THE GAP GETS

Estate-wide baselines miss the quiet services

7%

OF AROUND FIFTY MAJOR SERVICES, FAILING REGULARLY

Source: author's experience, a fifty-service estate; figures rounded

> FROM THE FIELD
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> It was the 500s that mattered. Across around fifty major services, about 7 per cent were failing regularly. Some were used ten or twenty times a day, some once a month, but each one sat inside a longer journey, so when it failed, the journey failed. One service had been throwing 500s under the radar for more than two weeks. No events on the floor, no nothing, because the anomaly detection had never fired.

Low-volume does not mean low-value. A service used twenty times a day can still be the step that decides whether a customer finishes what they came to do. Baselines learned across a whole estate will not notice it failing, because against the busy services it never looks unusual.

Three things to check in your own estate this month:

Which customer journeys depend on a service that is used only a few times a day or a few times a month?

For each of those services, is its normal learned from its own traffic, or from the estate around it?

If one of them had been failing for a fortnight, what would have told you, other than a complaint?

If the honest answer to the last one is "nothing", you are in the same position I was, and you will find out the same way.

AMAZON S3 · 28 FEBRUARY 2017

Your status view can fail with the system

On 28 February 2017, a member of Amazon's S3 team ran an established playbook command to remove a small number of servers from an S3 subsystem in Northern Virginia. One input was entered incorrectly and a larger set of servers was removed than intended.[3]

What followed is the outcome question at the largest scale most of us will ever see. AWS's own post-mortem records that from the start of the event until 11:37 AM Pacific time, it could not update the individual service statuses on its own Service Health Dashboard, because the dashboard's administration console depended on S3, the service that was down. It told customers through Twitter and banner text instead. Afterwards it moved that console to run across multiple regions.[3]

The lesson is not about Amazon. It is that the view you rely on to tell you what is happening can depend on the thing that is failing, and you only find out on the day. Ask of every leadership view you have: what does this depend on, and would it still tell me the truth if that failed?

METRICS & MAYHEM

THE FULL GUIDE

Get the full guide, free.

One question, a twenty-minute test to find your own gap, and what to do on Monday.

GET IT AT
masteringobservability.com/ebooks/outcome-question

Free. The link is in the first comment.

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